

Temporal Analysis and Short-Term Forecasting of Global Terrorism Incidents (1970–2017) Using the Global Terrorism Database

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Abstract—This study investigates the temporal dynamics of global terrorism incidents using the Global Terrorism Database (1970–2017). Monthly attack counts were constructed to analyze long-term trends, temporal variability, and potential seasonal patterns in terrorist activity. Exploratory temporal analysis was performed using time-series visualization, decomposition, and statistical diagnostics. Stationarity was evaluated using the Augmented Dickey–Fuller test, and temporal autocorrelation was examined through ACF and PACF analysis. Forecasting models were developed using ARIMA and linear regression approaches and evaluated using holdout validation. Among the tested models, an ARIMA(0,1,1) model applied to log-transformed attack counts achieved the lowest forecasting error and exhibited well-behaved residual diagnostics. In comparison, the linear regression model produced substantially higher prediction error, highlighting the importance of accounting for temporal dependence in terrorism data. The analysis demonstrates how time-series methods can reveal evolving patterns in global terrorist activity while also highlighting limitations in forecasting structural shifts in complex socio-political phenomena.

I. INTRODUCTION

Understanding the temporal dynamics of terrorist activity is important for identifying patterns in global security trends and for developing analytical tools that can support short-term forecasting of such events. Terrorism datasets often exhibit complex temporal behavior characterized by long-term trends, irregular fluctuations, and potential seasonal variations. Time-series analysis provides a systematic framework for examining such temporal structures, allowing researchers to study how event frequencies evolve over time and to build statistical models for forecasting future activity.

In this work, we analyze temporal patterns in global terrorist incidents using the Global Terrorism Database (GTD) [1], which records terrorist events worldwide from 1970 to 2017. The dataset contains detailed information about each recorded incident, including location, date, attack type, and casualty statistics. For the purpose of temporal analysis, the event-level data were aggregated into monthly attack counts, enabling the construction of a long-term time series spanning several decades. This representation allows for the examination of global terrorism trends, the identification of temporal

dependencies, and the application of statistical forecasting techniques.

The analysis begins with exploratory temporal data analysis, including visualization of long-term trends and seasonal components using time-series decomposition. Stationarity and temporal autocorrelation are then examined using statistical tests and diagnostic plots such as the Augmented Dickey–Fuller (ADF) test, autocorrelation function (ACF), and partial autocorrelation function (PACF). These analyses guide the selection of appropriate time-series models for forecasting.

To evaluate forecasting approaches, both autoregressive integrated moving average (ARIMA) models and a baseline linear regression model are applied to the monthly attack count series. The models are trained using historical observations and evaluated using holdout validation to assess their ability to predict short-term future activity. Model performance is compared using forecast error metrics, and diagnostic checks are used to evaluate residual behavior.

The objective of this study is twofold: first, to characterize the temporal behavior of global terrorism incidents using statistical time-series techniques, and second, to evaluate the effectiveness of different temporal forecasting models for short-term prediction. Through this analysis, we demonstrate how temporal statistical methods can provide insight into evolving patterns of global terrorist activity while also highlighting the limitations of purely historical models in capturing sudden structural changes in complex real-world phenomena. The scripts used for preprocessing the dataset, performing time-series analysis, and implementing forecasting models are available in the project repository [2].

II. DATASET DESCRIPTION

This study utilizes data from the Global Terrorism Database (GTD), an open-source dataset that contains detailed records of terrorist incidents worldwide. The dataset is maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism (START) and includes information on terrorist events from 1970 to 2017. Each entry in the dataset corresponds to a recorded terrorist incident and contains attributes

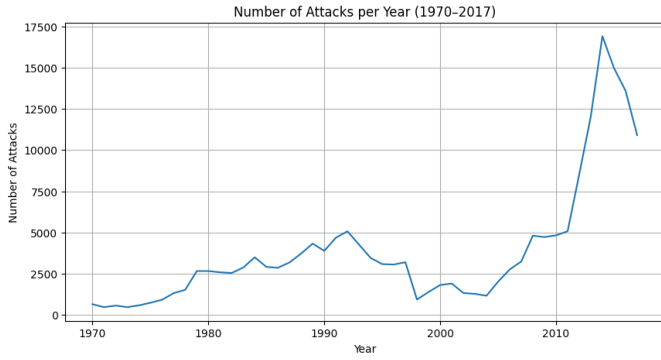


Fig. 1. Yearly number of terrorist incidents worldwide from 1970 to 2017.

describing the temporal, spatial, and operational characteristics of the event.

The GTD dataset includes variables such as the year, month, and day of the incident, geographical information (country, region, latitude, and longitude), attack type, target type, responsible group, and casualty statistics including the number of people killed and wounded. Because the focus of this assignment is temporal analysis, only the attributes related to event timing and casualties were used. Specifically, the variables *Year*, *Month*, *Day*, *Killed*, and *Wounded* were selected to construct the temporal dataset. To understand the long-term evolution of global terrorism incidents, the yearly number of attacks was first examined. Figure 1 shows the yearly distribution of terrorist incidents between 1970 and 2017, highlighting the gradual increase in activity leading up to the mid-2010s followed by a noticeable decline. The dataset contains more than 180,000 recorded terrorist incidents spanning nearly five decades, providing a sufficiently long temporal record for applying statistical time-series analysis methods.

Since the original dataset is event-based, where each row represents an individual incident, preprocessing was required to transform the data into a time-series format suitable for temporal analysis. The incident dates were converted into a standard datetime format and aggregated at a monthly resolution. Monthly aggregation was chosen to balance temporal granularity and stability of observations, as daily data can be highly sparse while yearly aggregation may obscure short-term patterns. The resulting time series represents the total number of terrorist incidents per month across the entire dataset period.

In addition to attack counts, two additional temporal variables were derived from the dataset: total fatalities and total casualties. Fatalities correspond to the total number of individuals killed in terrorist incidents during a given month, while casualties represent the combined number of individuals killed and wounded. These derived variables provide additional insight into the severity and impact of terrorist activity over time.

The resulting dataset forms a multivariate temporal representation of global terrorism activity from 1970 to 2017. The long temporal span and large number of recorded incidents

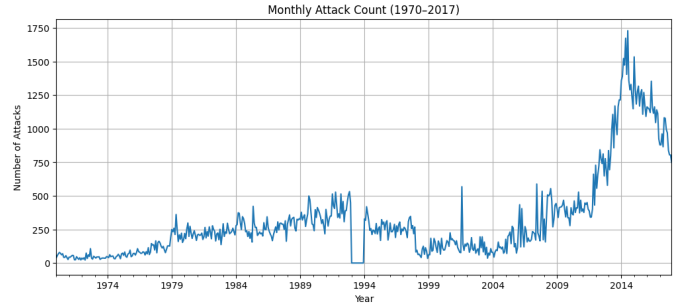


Fig. 2. Monthly global terrorist attack counts aggregated from the event-level dataset.

make the dataset suitable for time-series analysis techniques such as decomposition, stationarity testing, and forecasting using statistical models.

III. TEMPORAL DATA PREPROCESSING

Before applying forecasting models, the event-level dataset must be transformed into a structured time series suitable for temporal analysis. The Global Terrorism Database records individual terrorist incidents, where each row corresponds to a single event. Therefore, several preprocessing steps were performed to convert the event data into a consistent temporal representation suitable for time-series analysis.

A. Temporal Aggregation

The dataset contains temporal attributes including *Year*, *Month*, and *Day* for each recorded terrorist incident. These attributes were combined to construct a valid datetime representation for each event. Some entries contained missing or unspecified day values; these were replaced with a default value to allow consistent datetime construction.

After constructing the temporal index, the dataset was sorted chronologically and aggregated at a monthly resolution. Monthly aggregation was selected because daily observations are often sparse and highly irregular, while yearly aggregation may obscure short-term variations in terrorist activity. Aggregating the data at the monthly level provides a balanced temporal granularity while preserving meaningful patterns in the data.

The resulting time series represents the total number of terrorist incidents occurring globally in each month between 1970 and 2017. The resulting monthly attack count series used for temporal analysis is shown in Figure 2.

B. Time-Series Decomposition

To understand the underlying temporal structure of the data, classical time-series decomposition was applied to the monthly attack count series. Decomposition separates the observed time series into three components:

- A **trend component**, representing long-term changes in global terrorist activity,
- A **seasonal component**, representing recurring patterns within a yearly cycle, and

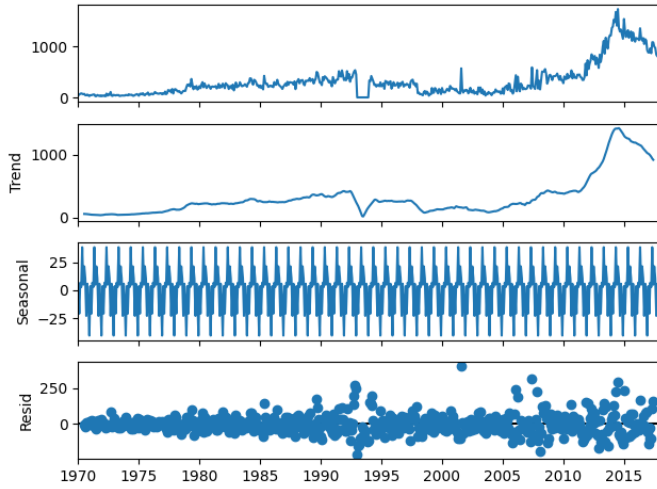


Fig. 3. Classical decomposition of the monthly attack count time series into trend, seasonal, and residual components.

- A **residual component**, representing irregular fluctuations not explained by the trend or seasonal components.

The decomposition revealed a strong long-term trend in global terrorist activity, particularly an increase in incidents leading up to the mid-2010s followed by a noticeable decline. The seasonal component was relatively small compared to the magnitude of the trend, suggesting that long-term structural changes dominate the temporal dynamics of the dataset. Figure 3 illustrates the decomposition of the monthly attack count series into trend, seasonal, and residual components.

C. Stationarity Preparation

Many time-series forecasting models require the underlying data to be stationary. Stationarity implies that the statistical properties of the time series, such as mean and variance, remain stable over time. Visual inspection of the monthly attack counts suggested the presence of a strong upward trend, indicating that the original series is non-stationary.

To address this issue, first-order differencing was applied to the time series. Differencing transforms the series into the change between consecutive observations, effectively removing long-term trends and stabilizing the mean of the series.

D. Variance Stabilization

In addition to trend-related non-stationarity, the variance of the attack count series increases over time as the magnitude of observations grows. To reduce heteroskedasticity and stabilize the variance, a logarithmic transformation was applied to the attack count series. The log transformation reduces the impact of extreme values and improves the stability of the time-series model during forecasting.

Through these preprocessing steps—temporal aggregation, decomposition, differencing, and variance stabilization—the dataset was prepared for subsequent autocorrelation analysis and forecasting using statistical time-series models.

IV. TEMPORAL AUTOCORRELATION AND STATIONARITY ANALYSIS

Understanding the temporal dependence structure of the data is an essential step in time-series modeling. In this study, statistical tests and diagnostic plots were used to examine the stationarity properties of the monthly attack count series and to identify appropriate model parameters for forecasting. In particular, stationarity testing, autocorrelation analysis, and partial autocorrelation analysis were conducted to guide the selection of ARIMA model components.

A. Stationarity Analysis

Many time-series models, including the autoregressive integrated moving average (ARIMA) [3] framework, assume that the underlying series is stationary. A stationary time series has statistical properties such as mean and variance that remain constant over time. Visual inspection of the monthly attack count series revealed a clear long-term trend and increasing variability, indicating that the original series is non-stationary.

To formally evaluate stationarity, the Augmented Dickey-Fuller (ADF) [4] test was applied to the monthly attack count series. The test evaluates the null hypothesis that the time series contains a unit root, which implies non-stationarity. The results of the ADF test indicated that the original series failed to reject the null hypothesis, confirming the presence of non-stationarity.

To address this issue, first-order differencing was applied to the time series. Differencing removes long-term trends by transforming the series into the change between consecutive observations. After applying first-order differencing, the ADF test produced a statistically significant result, indicating that the differenced series satisfies the stationarity assumption required for ARIMA modeling.

B. Autocorrelation Analysis

After achieving stationarity through differencing, the temporal dependence structure of the series was examined using the autocorrelation function (ACF). The ACF measures the correlation between observations separated by different time lags and helps identify potential moving average (MA) components in the data.

The ACF plot of the differenced series revealed a strong negative spike at lag 1 followed by correlations that rapidly decay within the confidence interval bounds. This pattern is characteristic of a moving average process of order one, suggesting that an MA(1) component may be appropriate for modeling the temporal dependence in the series. The autocorrelation structure of the differenced series is shown in Figure 4.

C. Partial Autocorrelation Analysis

In addition to the ACF, the partial autocorrelation function (PACF) was examined to assess the presence of autoregressive (AR) components. The PACF measures the correlation between observations at different lags after accounting for intermediate lag effects.

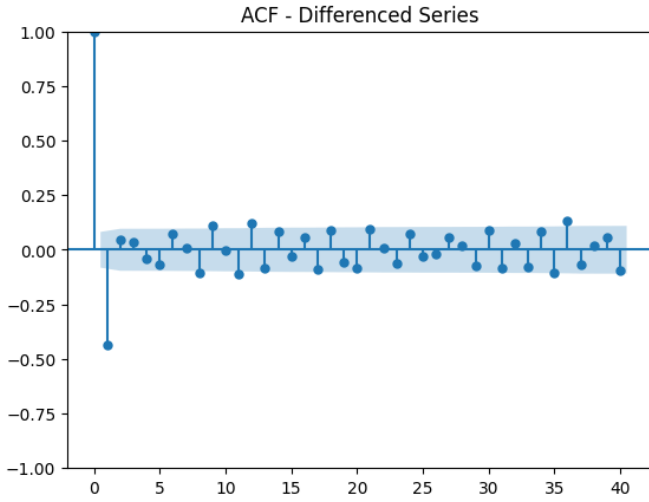


Fig. 4. Autocorrelation function (ACF) of the differenced monthly attack count series.

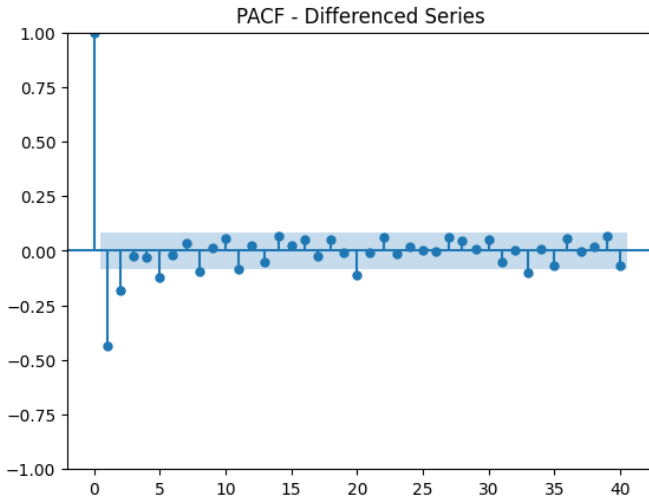


Fig. 5. Partial autocorrelation function (PACF) of the differenced monthly attack count series.

The PACF plot of the differenced series did not show a clear cutoff pattern at higher lags, and most correlations fell within the statistical confidence bounds. This behavior indicates that the series does not exhibit strong autoregressive structure, suggesting that an autoregressive component may not be necessary for the model. The partial autocorrelation structure of the differenced series is shown in Figure 5.

D. Implications for Model Selection

Based on the stationarity analysis and the observed autocorrelation structure, an ARIMA model with one order of differencing and a first-order moving average component was selected as the primary candidate for forecasting. This corresponds to an ARIMA(0,1,1) model. The model choice is supported by the strong lag-1 autocorrelation observed in the

differenced series and the absence of significant higher-order autoregressive behavior in the PACF.

This preliminary analysis provides the statistical justification for the ARIMA model structure used in the subsequent forecasting experiments.

V. TEMPORAL FORECASTING MODELS

After preprocessing and stationarity analysis, forecasting models were developed to predict short-term future terrorist activity. Two different modeling approaches were considered: an autoregressive integrated moving average (ARIMA) model, which captures temporal dependence in the data, and a baseline linear regression model, which models long-term trend behavior. The performance of these models was evaluated using holdout validation. Model selection was additionally supported by information criteria such as the Akaike Information Criterion (AIC), where lower values indicate better model fit while penalizing model complexity.

A. ARIMA Model

The autoregressive integrated moving average (ARIMA) model is a widely used statistical framework for time-series forecasting. ARIMA models represent temporal dependencies in a series through three components: autoregressive (AR) terms, differencing (I), and moving average (MA) terms. The general ARIMA(p, d, q) model includes:

- p : the number of autoregressive terms,
- d : the order of differencing required to achieve stationarity, and
- q : the number of moving average terms.

Based on the stationarity analysis conducted in the previous section, first-order differencing was sufficient to transform the monthly attack count series into a stationary process. Therefore, the differencing order was set to $d = 1$. The autocorrelation and partial autocorrelation analysis indicated a strong lag-1 moving average structure and minimal higher-order autoregressive effects. This pattern suggested that an ARIMA(0,1,1) model is appropriate for capturing the temporal dynamics of the dataset.

B. Log Transformation for Variance Stabilization

The monthly attack count series exhibits increasing variability over time, particularly during periods of high terrorist activity. To reduce heteroskedasticity and stabilize variance, a logarithmic transformation was applied to the attack count series prior to model estimation. The transformation helps compress large values and improves model stability when forecasting future observations.

Forecasts generated from the log-transformed model were subsequently transformed back to the original scale for evaluation and comparison with observed values.

C. Linear Regression Baseline

In addition to the ARIMA model, a simple linear regression model was implemented as a baseline forecasting approach. The regression model treats time as an explanatory variable

and estimates a linear relationship between time and the number of terrorist incidents. The model can be expressed as:

$$y_t = \beta_0 + \beta_1 t + \epsilon_t \quad (1)$$

where y_t represents the number of terrorist incidents at time t , β_0 and β_1 are regression coefficients, and ϵ_t represents the error term.

Unlike ARIMA models, linear regression does not explicitly account for temporal autocorrelation and therefore serves primarily as a baseline comparison for evaluating the effectiveness of time-series methods.

D. Holdout Validation Strategy

To evaluate forecasting performance, a holdout validation approach was used. The time series was divided into training and testing segments based on chronological order. Observations from 1970 to 2014 were used to train the models, while data from 2015 to 2017 were reserved for testing. This approach allows the models to be evaluated on unseen future observations, providing a realistic assessment of short-term forecasting performance.

Model performance was evaluated using the root mean squared error (RMSE) metric, which measures the average magnitude of prediction errors between the forecasted values and the observed test data. Lower RMSE values indicate better predictive performance.

VI. RESULTS AND DISCUSSION

This section evaluates the forecasting performance of the proposed models and analyzes the observed temporal behavior of global terrorism incidents. Model performance was assessed using the root mean squared error (RMSE) metric on the holdout test set covering the period from 2015 to 2017.

A. Forecasting Performance

The forecasting accuracy of the models was evaluated using RMSE on the test data. Table I summarizes the forecasting performance of the evaluated models.

TABLE I
FORECASTING PERFORMANCE COMPARISON

Model	RMSE
Linear Regression	600.08
ARIMA(0,1,1) (Raw Series)	213.03
ARIMA(0,1,1) (Log-Transformed Series)	190.83

The results indicate that the ARIMA model significantly outperforms the linear regression baseline. The linear regression model produced the largest prediction error, as it assumes a simple linear relationship between time and attack counts and does not account for temporal autocorrelation. In contrast, the ARIMA model captures short-term temporal dependencies, allowing it to produce more accurate forecasts.

Applying a logarithmic transformation to the attack count series further improved forecasting performance. The log-transformed ARIMA model achieved the lowest RMSE among

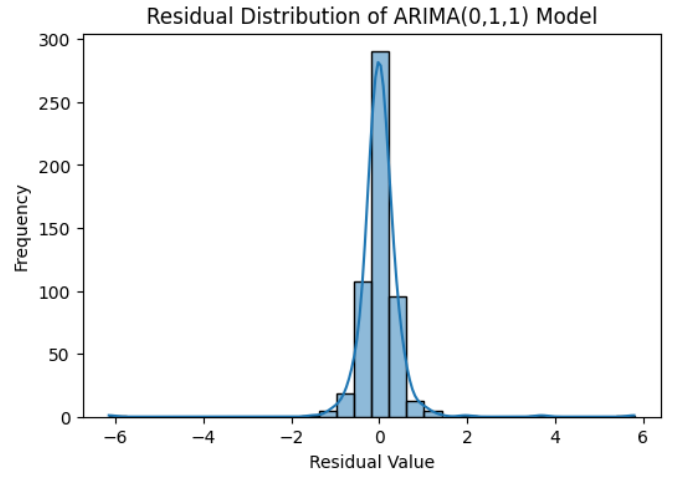


Fig. 6. Histogram of residuals from the ARIMA(0,1,1) model. The residuals are approximately centered around zero, suggesting that the model adequately captures the temporal structure of the series.

the evaluated models, indicating that variance stabilization helped improve predictive accuracy. To further evaluate model adequacy, the distribution of residuals was examined. The log-transformed ARIMA model also produced the lowest AIC value among the evaluated models, further supporting its suitability for forecasting the attack count series. The histogram of residuals (Figure 6) shows that the residual values are approximately centered around zero with no strong skewness, suggesting that the ARIMA model captures the primary temporal structure of the series.

B. Forecast Behavior

Figure 7 compares the predicted values from the ARIMA model with the observed attack counts in the test period. The ARIMA model captures the overall magnitude of the time series but produces relatively stable forecasts over longer horizons. This behavior is expected for ARIMA models with differencing, which tend to produce forecasts that converge toward a constant level when no strong autoregressive structure is present.

Although the model captures general levels of activity, it does not fully reproduce the sharp decline in terrorist incidents observed after 2015. This discrepancy suggests the presence of structural changes in the underlying process governing terrorist activity.

C. Model Limitations

The forecasting results highlight several limitations of purely historical time-series models. Terrorist activity is influenced by complex geopolitical, economic, and social factors that may change abruptly over time. As a result, models that rely only on past observations may struggle to anticipate sudden structural shifts in the data.

In the present analysis, the training data contained a period of increasing terrorist activity leading up to 2014, whereas the test period experienced a notable decline in attack counts.

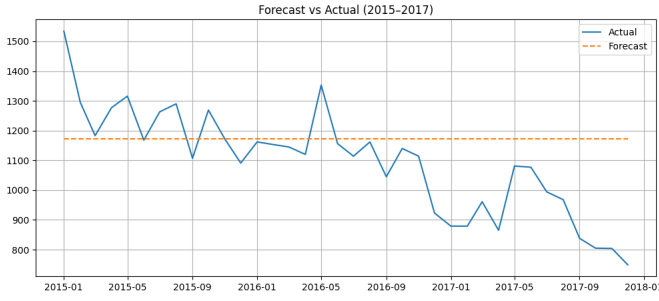


Fig. 7. Comparison of ARIMA forecast and observed attack counts during the holdout test period (2015–2017).

Since ARIMA models extrapolate patterns observed in historical data, the model was unable to fully anticipate this regime change.

Despite these limitations, the ARIMA model still provides valuable insight into the short-term temporal dynamics of terrorist activity and significantly improves predictive accuracy compared to simple trend-based models.

D. Implications for Temporal Analysis

The results demonstrate that incorporating temporal dependence through ARIMA modeling leads to substantial improvements in forecasting performance compared to simple regression approaches. The findings also highlight the importance of proper preprocessing, stationarity analysis, and variance stabilization when applying time-series models to complex real-world datasets.

Overall, the analysis shows that time-series techniques are effective tools for exploring temporal patterns in large-scale event datasets such as the Global Terrorism Database. However, forecasting accuracy may be limited when sudden structural changes occur in the underlying data-generating process.

VII. CONCLUSION

This study investigated the temporal dynamics of global terrorist incidents using the Global Terrorism Database covering the period from 1970 to 2017. Event-level data were aggregated into a monthly time series to enable temporal analysis and forecasting. Exploratory time-series techniques, including decomposition, stationarity testing, and autocorrelation analysis, were used to understand the underlying structure of the data and to guide model selection.

Based on the observed autocorrelation patterns and stationarity analysis, an ARIMA modeling framework was applied to forecast short-term terrorist activity. Among the evaluated models, an ARIMA(0,1,1) model applied to a log-transformed attack count series achieved the best forecasting performance. The results demonstrated that accounting for temporal dependence significantly improves predictive accuracy compared to a simple linear regression model. Residual diagnostics confirmed that the selected ARIMA model adequately captured the autocorrelation structure of the data.

Although the ARIMA model performed well in terms of prediction error, the analysis also revealed limitations of purely historical forecasting approaches. In particular, the model was unable to fully anticipate the sharp decline in global terrorist incidents observed after 2015, highlighting the presence of structural shifts that cannot be captured by simple univariate time-series models.

Future work could extend this analysis by incorporating spatial information, regional-level modeling, or exogenous variables such as geopolitical indicators to better capture the complex drivers of terrorist activity. Such approaches may improve forecasting performance and provide deeper insight into the evolving dynamics of global terrorism.

DECLARATION OF GENERATIVE AI TOOLS

Generative AI tools were used during the preparation of this assignment to assist with code debugging, clarification of time-series analysis concepts, and improvement of the structure and clarity of written explanations. These tools were also used to review sections of the report for grammatical correctness and readability.

All data preprocessing steps, implementation of the temporal analysis pipeline, model development (including ARIMA and regression models), and interpretation of the results were performed and verified by the authors. The final code, figures, experimental results, and conclusions presented in this report reflect the authors' understanding of temporal statistical analysis applied to the Global Terrorism Database.

TASK OWNERSHIP DECLARATION

This project was carried out collaboratively by the team members Keshav, Himanshu, and Rishit. Each member contributed to different aspects of the data analysis, model implementation, and interpretation of results.

Keshav was primarily responsible for the core temporal analysis pipeline. This included preprocessing the Global Terrorism Database into a monthly time-series format, performing time-series decomposition and stationarity analysis, implementing ARIMA forecasting models, and conducting model validation using holdout testing and error metrics.

Himanshu contributed to the exploratory analysis of the dataset and the generation of temporal visualizations used throughout the study. This included preparing trend plots, supporting the interpretation of temporal patterns in the data, and assisting in examining decomposition and autocorrelation behavior.

Rishit contributed to the implementation and evaluation of baseline models used for comparison with the ARIMA approach. This included assisting with the linear regression model, organizing experimental outputs, and supporting the comparison of model performance.

All team members jointly participated in interpreting the results and preparing the final report.

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